

Unit 2: Tidyverse & Ggplot Review

1 What is Tidyverse?

The tidyverse package is a collection of libraries in R that streamline data manipulation. Some of the most common functions we used during labs include:

- **tibble()**
 - Constructs an enhanced data frame called a tibble
- **read_csv()**
 - Like read.csv() but returns a tibble and functions faster
- **mutate()**
 - Used to create new or edit previous variables/columns
- **filter()**
 - Can select rows that specify certain conditions
- **group_by()**
 - For specifying grouped variables for data operations
- **|>**
 - A “pipe” which is used to make code easier to read by feeding previous operations to the next line
- **case_when()**
 - When multiple if_else() conditions need to be checked
- **summarize()**
 - Allows to ask for statistics calculated on selected columns of a data frame
- **separate()**
 - Used to separate a column into multiple columns using a delimiter

Why use Tidyverse over Base R?

Tidyverse notation is easier to read and write compared to base R. Using pipes, |>, steps flow from line to line, making the code easier to follow and understand, which also makes comments more comprehensible because you can add descriptions to each step as you go.

2 What is Ggplot?

Ggplot2 is a data visualization package in tidyverse used to create clear, customizable graphs. It is based on layers, where you build plots step by step by adding components like data, axes, and visual elements.

Things to keep in mind:

- What shape will the data follow, if any
- Put the plot in a figure environment so it is nicely formatted. We will use a couple of parameters for the environment:
 - label: How we can refer to the figure later in the file
 - fig-cap: What the title will show up as
 - fig-width: how many inches wide the plot will be
 - fig-height: how many inches high the plot will be
 - fig-align: where relative to the page the figure will appear

Let's simulate random data following the normal distribution to work with. The seed will ensure it does not change.

```
1 set.seed(61)
2 data = data.frame(
```

```

3   var1 = rnorm(100),
4   var2 = rnorm(100)
5 )

```

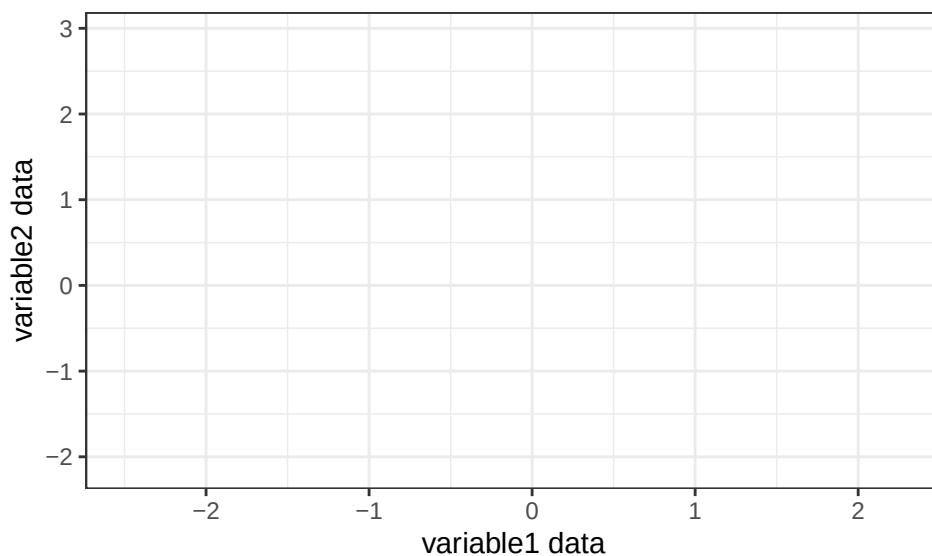
First you tell your plot what data to use. I added `theme_bw()` and `labs()` to clean up the graph by making it black and white with custom axis labels:

```

1 library(ggplot2)
2 ggplot(data, aes(x = var1, y = var2)) +
3   labs(x = "variable1 data",
4        y = "variable2 data") +
5   theme_bw()

```

Figure 1: Empty Plot



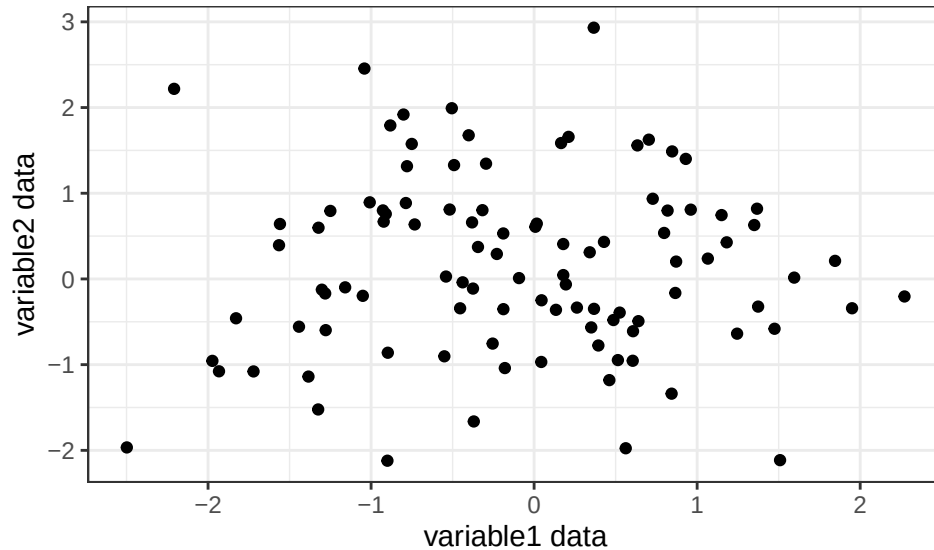
Then use `geom` to actually add a shape to the plot:

```

1 ggplot(data, aes(x = var1, y = var2)) +
2   geom_point() +
3   labs(x = "variable1 data",
4        y = "variable2 data") +
5   theme_bw()

```

Figure 2: Scatter Plot of Data



One of the best uses for ggplot is data summarization, a key step in data analysis once you have cleaned the data set using tidyverse. One of the most important decisions is choosing which geom is right for your data. One way is to check what variables your data consists of.

1 Categorical Variable

Go to plots would be either a bar plot or a doughnut plot.

- `geom_bar()`
- `coord_polar()`

1 Quantitative Variable

Go to plots are a histogram, violin plot, and box plot. Helpful to overlay the violin plot on a box plot.

- `geom_histogram()`
- `geom_violin()`
- `geom_boxplot()`

2 Categorical Variables

Go to plots are the mosaic, grouped bar, and grouped doughnut plots.

- `ggmosaic::geom_mosaic()`
- `geom_bar()`
- `coord_polar()`

2 Quantitative Variables

- `geom_point()`

1 Categorical and 1 Quantitative Variable Go to plots are grouped histogram, grouped violin, and grouped box plots. Helpful to overlay a box plot on a violin plot.

- `geom_histogram()`
- `geom_violin()`
- `geom_boxplot()`

3 Plotting Data

Let's see these plots in action. But first, we will need some data.

```

1 set.seed(61)
2 n = 200
3
4 data = data.frame(
5   # One quantitative variable
6   score = rnorm(n, mean = 75, sd = 10),
7
8   # Two quantitative variables
9   study_hours = runif(n, 0, 10),
10  exam_grade = 50 + 4 * runif(n, 0, 10) + rnorm(n, sd = 5),
11
12  # One categorical variable
13  genre = sample(c("Rock", "Pop", "Jazz", "Classical"), n, replace = TRUE),
14
15  # Two categorical variables
16  grade_level = sample(c("Freshman", "Sophomore", "Junior", "Senior"), n,
17                      replace = TRUE), pass_fail = sample(c("Pass", "Fail"),
18                  n, replace = TRUE, prob = c(0.75, 0.25)),
19
20  # One categorical variable and one quantitative variable
21  subject = sample(c("Math", "English", "Science", "History"), n, replace = TRUE)
22 )

```

Now let's make the graphs in order.

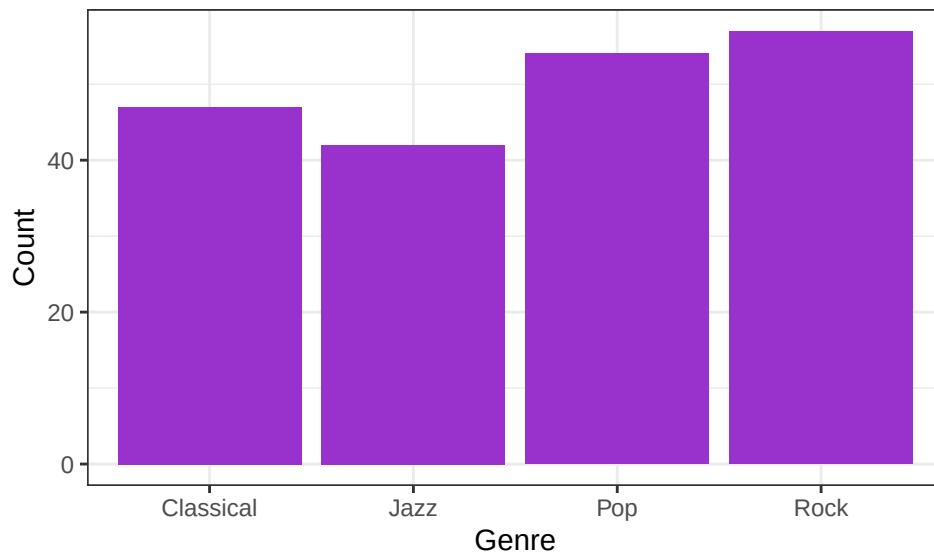
This is a Bar plot for data with one categorical variable!

```

1 # Bar Plot
2 ggplot(data, aes(x = genre)) +
3   geom_bar(fill = "darkorchid") +
4   labs(x = "Genre", y = "Count") +
5   theme_bw()

```

Figure 3: Bar Plot



These are the histogram, violin, and box plots for data with one quantitative variable!

```

1 # Histogram
2 ggplot(data, aes(x = score)) +

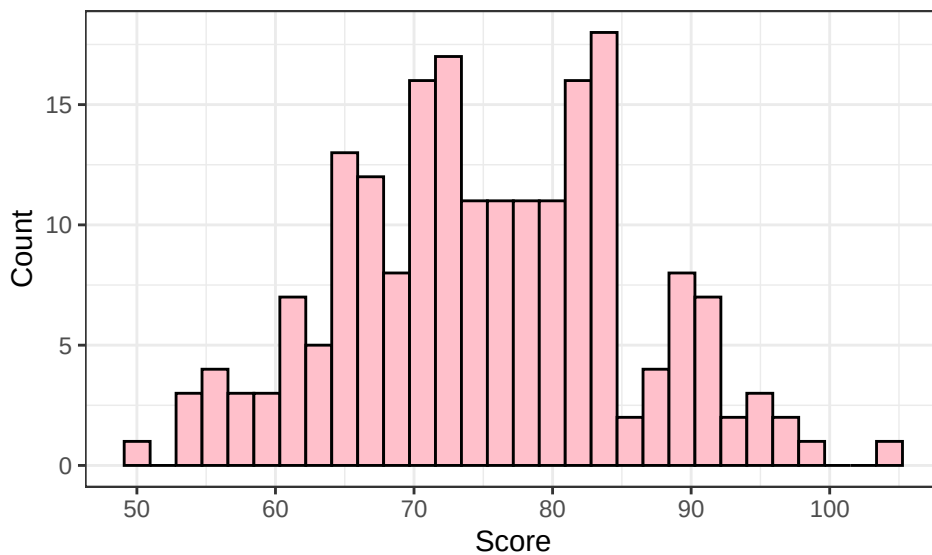
```

```

3 geom_histogram(fill = "pink", color = "black") +
4 labs(x = "Score", y = "Count") +
5 theme_bw()

```

Figure 4: Histogram Plot

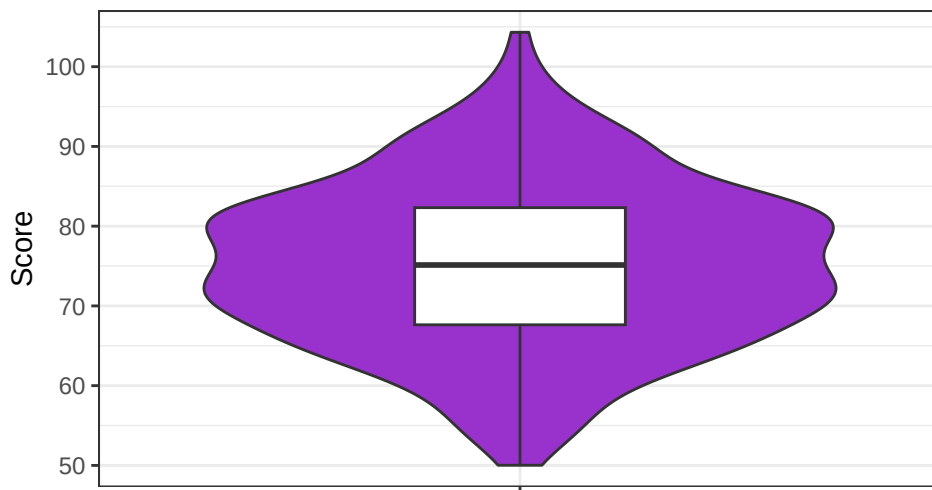


```

1 # Violin Plot and box plot overlay
2 ggplot(data, aes(x = "", y = score)) +
3   geom_violin(fill = "darkorchid") +
4   geom_boxplot(width = .3) +
5   labs(x = "", y = "Score") +
6   theme_bw()

```

Figure 5: Violin Plot and Box Plot Overlay



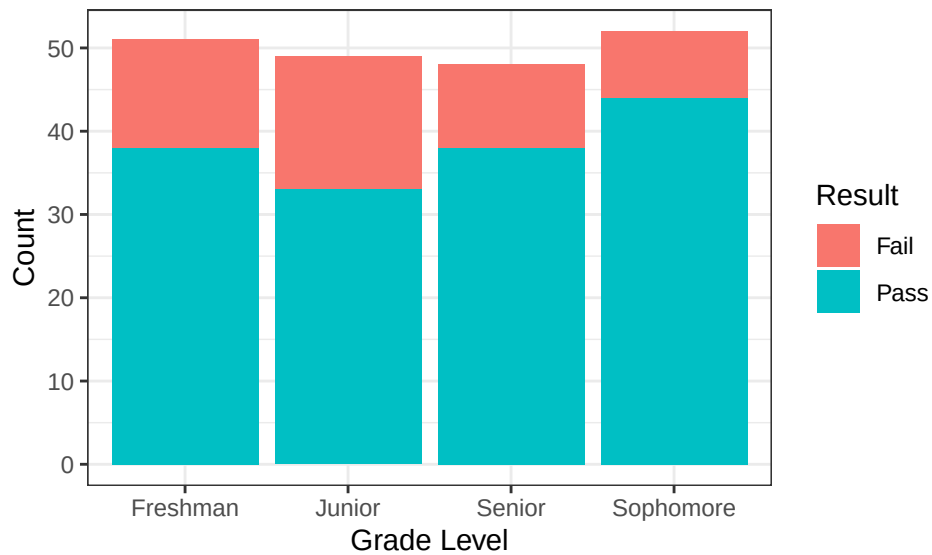
Here is the grouped bar plot for data with 2 categorical variables!

```

1 # Grouped Bar Plot
2 ggplot(data, aes(x = grade_level, fill = pass_fail)) +
3   geom_bar() +
4   labs(x = "Grade Level", y = "Count", fill = "Result") +
5   theme_bw()

```

Figure 6: Grouped Bar Plot



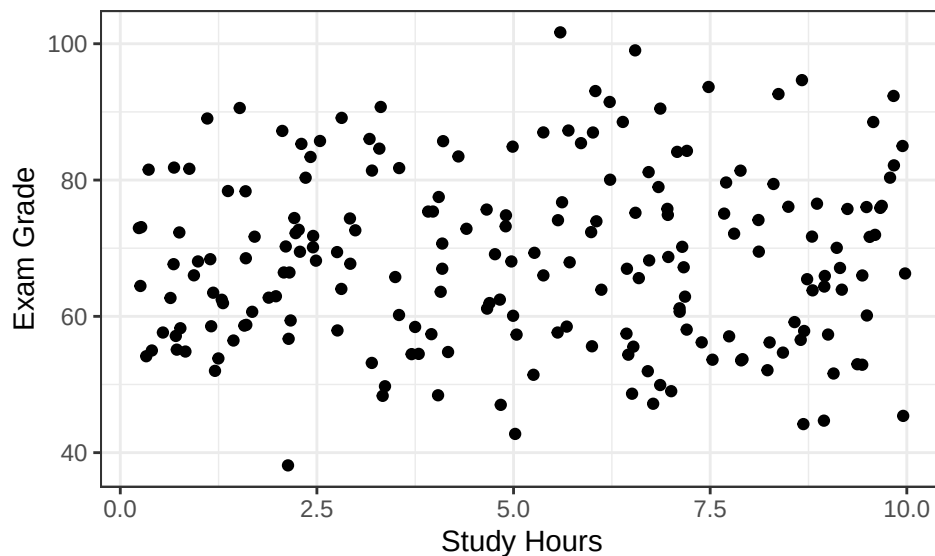
Here is a scatterplot for data with one quantitative variable!

```

1 ggplot(data, aes(x = study_hours, y = exam_grade)) +
2   geom_point() +
3   labs(x = "Study Hours", y = "Exam Grade") +
4   theme_bw()

```

Figure 7: Scatter Plot of Data

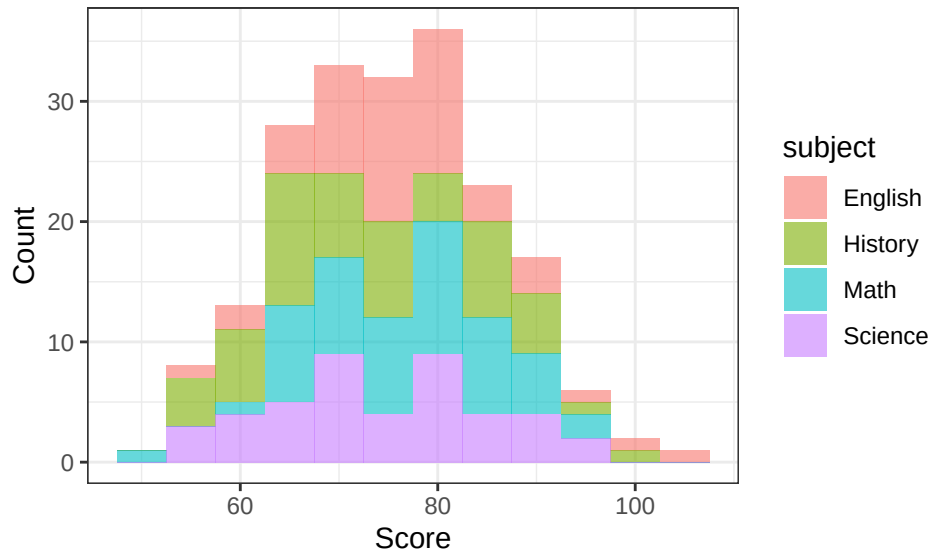


Here are grouped histogram, violin, and box plots for data with one quantitative variable and one

categorical variable!

```
1 # Grouped Histogram
2 ggplot(data, aes(x = score, fill = subject)) +
3   geom_histogram(binwidth = 5, alpha = 0.6) +
4   labs(x = "Score", y = "Count") +
5   theme_bw()
```

Figure 8: Grouped Histogram



```
1 # Grouped Violin Plot
2 ggplot(data, aes(x = subject, y = score, fill = subject)) +
3   geom_violin(alpha = 0.7) +
4   geom_boxplot(width = .3) +
5   labs(x = "Subject", y = "Score") +
6   theme_bw()
```

Figure 9: Grouped Violin Plot with Grouped Box Plot

